CSCN8020 ASSIGNMENT 3

Deep Q-Network Control of the Unitree G1 Left Elbow

Student full name	Emmanuel Ihejiamaizu
Student ID	9080005
Course	CSCN8020 - Reinforcement Learning
Assignment	Assignment 3

Project Summary

This project implements a student-written Deep Q-Network in PyTorch to control the left elbow of a fixed-base Unitree G1 humanoid robot in MuJoCo. The agent receives a four-value observation and selects one of three discrete actions to move toward target angles between -0.8 and +0.8 radians. Two epsilon-decay configurations were trained headlessly on CPU and evaluated greedily across four benchmark goals. Both achieved 100% success over 20 evaluation episodes. Configuration A was selected because it produced the highest mean evaluation reward, shortest mean episode length, and lowest final error. The repository includes source code, checkpoints, metrics, plots, video, notebook, README, and technical report.

GitHub Repository URL

https://github.com/chooksemmanuel/CSCN8020_Assignment3

Cloneable .git URL

https://github.com/chooksemmanuel/CSCN8020_Assignment3.git

Public repository prepared for grading and reproducibility review.